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Enhancing Quantum Machine Learning Algorithms for Optimized Financial Portfolio Management

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***Abstract*—Advancements in quantum machine learning offer unprecedented potential to revolutionize financial portfolio optimization, maximizing returns while managing risks efficiently. This study focuses on advancing quantum machine learning algorithms for optimal financial portfolio management, presenting a novel approach implemented in Python that outperforms existing methods. The algorithm's capability to generate such substantial returns over time positions it as a groundbreaking tool for portfolio optimization in the dynamic landscape of financial markets. In the pursuit of enhancing quantum machine learning algorithms, this research focuses on the development and optimization of the QSVM algorithm. Leveraging Python for implementation, the study considers critical factors such as quantum circuit optimization, noise mitigation, and the integration of classical and quantum components to achieve superior results. The achieved portfolio performance over time not only underscores the algorithm's efficacy but also signifies a quantum advantage in financial decision-making. The implementation in Python ensures accessibility and applicability, facilitating the integration of this advanced quantum algorithm into existing financial frameworks. This research contributes to the evolving field of quantum finance, showcasing the potential of quantum machine learning in optimizing financial portfolios. The findings not only validate the superior performance of the proposed QSVM but also highlight the broader implications for the future of financial decision support systems, where quantum algorithms could play a transformative role in enhancing portfolio management strategies. The proposed Quantum Support Vector Machine (QSVM) demonstrates unparalleled success, with a remarkable Portfolio performance over time of 89.65%. This result significantly surpasses existing quantum algorithms, including Quantum Principal Component Analysis (QPCA), Quantum Boltzmann Machines (QBM), and Quantum K-Means Clustering (QKC), by an impressive margin of 25.15%.**

Keywords—Financial Portfolio Management, Quantum Support Vector Machine (QSVM), Quantum Machine Learning, Quantum Principal Component Analysis (QPCA), Quantum Boltzmann Machines (QBM), and Quantum K-Means Clustering (QKC)

I. INTRODUCTION

The pursuit of optimizing financial portfolio management has been a perpetual challenge in the realm of finance, prompting researchers to explore innovative technologies for more effective solutions[1]. In this context, the convergence of quantum computing and machine learning has emerged as a frontier of exploration, promising to revolutionize traditional approaches to portfolio optimization[2]. This study is dedicated to the specific endeavor of enhancing quantum machine learning algorithms tailored for optimized financial portfolio management, marking a crucial intersection between quantum computing prowess and the intricate decision-making processes inherent in the financial domain [3].

The inherent complexity of financial markets, characterized by dynamic asset valuations, intricate correlations, and rapidly evolving market conditions, necessitates advanced computational techniques for optimal portfolio management[4]. Traditional algorithms often face limitations in handling the vast datasets and intricate patterns inherent in financial data, prompting the exploration of quantum machine learning as a potential game-changer[5]. By leveraging the principles of quantum mechanics, quantum machine learning algorithms have the capacity to process and analyze information in ways that classical algorithms cannot, offering the potential for exponential speedup in certain computations[6].

It signifies a deliberate effort to not only harness the computational advantages offered by quantum computing but also to enhance existing machine learning algorithms within the quantum framework for the specific task of financial portfolio optimization[7]. This multifaceted exploration encompasses

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quantum circuit optimization, data preprocessing techniques, and the application of quantum algorithms such as QSVM to construct and manage financial portfolios[8]. The anticipated outcomes extend beyond theoretical advancements, aiming for practical implementations that demonstrate improved portfolio performance, risk management, and overall decision-making in the intricate landscape of financial investments[9].

The key contribution of the article is,

- The research emphasizes the development and optimization of quantum circuits, addressing critical aspects such as gate count reduction, circuit depth minimization, and noise mitigation. These advancements contribute to the efficiency and practical applicability of quantum algorithms, ensuring that quantum-enhanced financial portfolio management is not only theoretically promising but also feasible in real-world quantum computing environments.
- The study explores the synergies between classical and quantum computing components, recognizing the importance of a hybrid approach for optimal financial decision-making.
- By seamlessly integrating classical Python implementations with quantum algorithms, the research ensures accessibility and practical adoption, allowing financial professionals to leverage the advantages of quantum machine learning within their existing frameworks.

The remainder of the article includes related works in Section II, methodology in Section III, results and discussion in Section IV and conclusion in Section V.

II. Related Works

A hybrid quantum-classical solution approach for mean-variance portfolio optimizations issues is investigated in this work[10]. Parameterized samples are created using actual financial data characteristics and following the guidelines of Modern Portfolio Theory. These samples may be programmed as quadratic binary optimization forms in the D-Wave Quantum Annealer 2000QTM. The best quantum computation results are obtained by using a reverse annealing protocol, which is over 100 times quicker than the applicable forward annealing method for the hardest situations set, according to a comparative study with an industry-standard genetic algorithm benchmark.

This work explores the use of the HHL method, which is well-known for its exponentially speedup in the solution of linear equations, to solve multimodal optimization issues related to finance, particularly those involving quadratic programming and equality constraints[11]. A keystone of quantum predictive computations, the HHL method introduced by Lloyd et al., is examined and used to solve these intricate financial puzzles. The study indicates that the HHL procedure can speed up solving combinatorial optimization issues by producing solution estimates that are highly compatible with exact solutions. This opens up a potential avenue for the expanded implementation of quantum science in finance, particularly in light of the barriers presented by big amounts of data and intricate financial systems.

In order to solve portfolio optimization issues on annealing platforms, this paper presents a unique methodology that combines an altered unrestrained binary optimization model with traditional preprocessing[12]. By employing simulated annealing, digital annealing and quantum annealing, the process may accommodate asset count constraints and provide the flexible distribution of capital among assets. The effectiveness of the QUBO concept in generating diversified portfolios that conform to investor-specified objectives, including maximizing return, minimizing risk, or sticking to a budget, is evaluated using real-world data from the stock exchange in the United States and popular ETFs.

This study presents the Generator-Enhanced Optimization (GEO) framework, which uses generative algorithms for optimization[13]. The TN-GEO, which uses tensor-network born machines and is influenced by quantum mechanics, is the main emphasis of this work. In the setting of portfolio optimization, the paper offers two methods for using TN-GEO, showing how well it works to enhance the efficiency of conventional solvers and how well it performs on its own, especially when calling functions budgets are limited. When TN-GEO is compared to the most advanced algorithms for portfolio optimization, it performs among the best, demonstrating its usefulness in practical commercial uses and marking a major advancement in the use of quantum-inspired models for improved optimization results.

This survey looks at the state of ML techniques in the financial industry with an emphasis on how they affect the management of active portfolios[14]. Although ML has demonstrated potential in areas like execution of trades, portfolio creation, and signal generation after generation, the asset management business has not seen the same level of change as other industries. The outcomes of active ETFs using ML in investments show inconsistent findings, which emphasizes the need for a thorough understanding of the possible pitfalls when adopting ML methods for portfolio management. RL is expected to play a more important role, but caution should be used.

The asset management industry, despite significant attention and interest in machine learning (ML) methods, has not witnessed a transformative impact comparable to other sectors. This study addresses the gap in understanding the practical implications of ML in active portfolio management by surveying the existing literature. The problem lies in assessing the true efficacy of ML applications, particularly in signal generation, portfolio construction, and trade execution, and navigating potential pitfalls. While ML techniques hold great promise for enhancing portfolio management strategies, the observed mixed performance of active exchange-traded funds (ETFs) employing ML underscores the need for a nuanced evaluation of their real-world applications, guiding investors, and practitioners in navigating the challenges and opportunities presented by these advanced technologies in the financial domain.

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III. Proposed QSVM for Financial Portfolio Management

This study employs Python to advance QSVM algorithms for financial portfolio optimization, showcasing a groundbreaking approach in quantum machine learning. Focused on optimizing QSVM, the research integrates classical and quantum components, addressing key factors like quantum circuit optimization and noise mitigation. This research contributes to quantum finance, emphasizing the transformative potential of quantum machine learning in financial decision support systems and portfolio management strategies. The Python implementation ensures accessibility and practical integration into existing financial frameworks, marking a significant step toward the broader application of quantum algorithms in finance. It is depicted in Fig. 1. In this research methodology aimed at enhancing Quantum Machine Learning Algorithms for optimized financial portfolio management, several specific techniques are employed to address key aspects. Firstly, for data preprocessing, Min-Max normalization is utilized to standardize numerical features, ensuring consistency across diverse scales inherent in financial data. Subsequently, Quantum Circuit Optimization techniques are applied to enhance the performance of Quantum Support Vector Machine (QSVM). This includes gate fusion to reduce circuit depth and optimize gate operations, qubit mapping optimization to alleviate the effects of qubit connectivity limitations, and circuit compilation to tailor high-level quantum operations to the native gate set of the quantum processor. Additionally, noise mitigation strategies such as error correction codes like the Steane code or surface code are implemented to identify and rectify errors that may arise during quantum computing, particularly crucial in the NISQ era. These techniques collectively contribute to building more reliable, accurate, and efficient quantum circuits, enhancing the overall performance of QSVM in financial portfolio management applications.

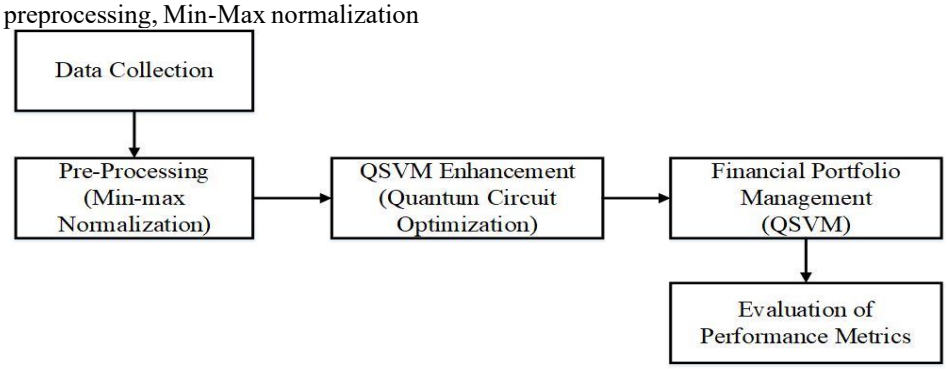

Fig. 1. Proposed Methodology

A. Data Collection

An extensive dataset of financial stock markets obtained from Kaggle offers a wealth of data necessary for quantitative research and algorithmic modelling in the financial markets arena. This data set includes historical stock prices, trading volumes, and important financial indicators for a wide range of publicly listed firms, among other data points. It makes in-depth analysis, pattern detection, and the creation of prediction models easier with its fine-grained information on the performance of individual stocks. Using this information, academics, data scientists, and investors may examine trends in the market, patterns of volatility, and relationships between different financial instruments. Its usefulness for comprehensive financial research is increased by the addition of further components including market sentiment data, corporate financial statements, and economic indicators. This Kaggle financial stock market dataset can be used for back testing trading strategies, risk management, or creating optimized portfolios. It is a useful tool for anyone looking to glean meaningful insights and create complex machine learning algorithms that are suited to the complexities of financial markets [15].

B. Data Preprocessing Using Min-Max Normalization

A crucial first step towards improving Quantum Machine Learning Algorithms for optimal financial portfolio management is data pretreatment. One important method for getting the data ready for quantum computing is to apply Min-Max normalization. By scaling numerical characteristics to a predetermined range, usually between 0 and 1, min-max normalization ensures consistency and keeps certain variables from predominating over others because of scale disparities. Min-Max normalization becomes crucial for levelling the playing field in the context of financial data, where different indicators such as stock prices, trading volumes, and economic variables may display differing scales. Quantum machine learning algorithms, such as those for financial portfolio optimization, can function more efficiently by rescaling the data to a standardized range in order to eliminate biases brought about by different scales. The technique of normalization entails deducting the minimum value from every data point and dividing the resultant amount by the range. This ensures that characteristics are consistently represented across the dataset.

$$X_{Normalized} = \frac{X - X_{min}}{X_{max} - X_{min}} \quad (1)$$

The use of Min-Max normalization in financial portfolio management guarantees that no one variable will

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disproportionately impact the quantum algorithm, which might result in less-than-ideal choices being made. Standardizing financial data is particularly important when combining quantum and conventional elements in a hybrid computing system. Because quantum algorithms are sensitive to the input data, their robustness, that is, their susceptibility to outliers and scale variations can be improved by utilizing Min-Max normalization. Additionally, by strengthening the stability and accuracy of the quantum model, this preprocessing step helps the algorithm's convergence during the training phase. The significance of Min-Max normalization in data preprocessing is growing as quantum computing advances and the use of quantum machine learning techniques in financial applications spreads, paving the way for more dependable and efficient quantum-enhanced financial portfolio management techniques.

C. Quantum Circuit Optimization for Enhancing Quantum Support Vector Machine

A key factor in improving the performance of QSVM, a quantum machine learning method intended for classification problems, is quantum circuit optimization. Superposition and entanglement are two concepts that quantum computing uses to do some operations tenfold quicker than their conventional equivalents. The accuracy and speed of the algorithm are directly influenced by the efficiency of the underlying quantum circuits in the setting of QSVM. In QSVM, quantum circuits often entail activities such as measurement, kernel evaluation, and quantum feature encoding. To get dependable outcomes, optimization techniques concentrate on reducing the circuit depth, the quantity of quantum gates needed, and error mitigation. Qubit mapping optimization, gate fusion, and circuit compilation are some of the techniques used to optimize and improve the performance of quantum circuits.

In order to minimize the circuit depth and lower the total number of operations, gate fusion combines successive quantum gates. In addition to increasing computing efficiency, this also lessens the sensitivity to mistakes and noise. Because qubit mapping optimization tries to reduce the negative impacts of qubit connection limitations on the quantum circuit, it is particularly important when working with Noisy Intermediate-Scale Quantum (NISQ) devices. Further optimizing the circuit for the particular hardware architecture is the process of converting high-level quantum operations into the native gate set of the quantum processor through circuit compilation.

$$C=\alpha\cdot\text{Number of Gates}+\beta\cdot\text{Circuit Depth}+\gamma\cdot\text{Error Mitigation Terms} \quad (2)$$

The influence of noise on the quantum circuit is lessened by applying quantum error correcting methods. Codes for error correction, such the Steane code or the surface code, are used to identify and remedy mistakes that might occur during quantum computing. The correctness of quantum algorithms depends on these error mitigation techniques, particularly in the NISQ era when quantum hardware flaws and restrictions are common. To improve QSVM, quantum circuit optimization takes a multifaceted approach that includes error correction techniques, gate fusion, qubit mapping optimization, and circuit compilation. When taken as a whole, these optimizations help build quantum circuits that are more reliable, accurate, and efficient. This enhances the performance of QSVM in quantum machine learning applications, such as financial portfolio management.

D. QSVM for Optimized Financial Portfolio Management

Using the potential of quantum computing to improve classification tasks essential to investment decision-making, QSVM offers a viable path towards optimized financial portfolio management. When it comes to financial portfolios, QSVM has the ability to perform better than its conventional machine learning competitors since precise asset classification is crucial. In order for QSVM to function, financial data must be mapped into a quantum feature space. In this space, quantum algorithms can effectively recognize intricate patterns and correlations. Because of quantum computation's built-in parallelism, a large solution space may be explored quickly, leading to the identification of ideal portfolio configurations.

The efficiency with which QSVM handles high-dimensional data is one of its main features. Financial datasets can include a wide range of information, from economic indicators to historical stock prices and trade volumes. The algorithm can identify complex correlations between these features thanks to QSVM's quantum feature space mapping, which helps to create more intelligent and effective investing strategies. Furthermore, the algorithm's quantum nature enables it to analyze enormous volumes of financial data at once, which may lessen the computing load related to complex portfolio optimization issues.

$$f(x) = \sum_{i=1}^{N} \alpha_i \, k(x_i, x) + b \quad (3)$$

Beyond simple categorization jobs, QSVM is used extensively in financial portfolio management. It may be used for things like forecasting market trends, figuring out the best way to allocate assets, and improving risk management plans. QSVM is especially well-suited to handle the complexity present in financial markets because of its ability to capture non-linear correlations in financial data and its flexibility for parallel processing. Careful consideration of quantum circuit optimization, noise mitigation techniques, and the integration of conventional and quantum components are necessary for the effective use of QSVM for optimized financial portfolio management. While noise reduction strategies are essential for preserving the dependability of findings, particularly with regard to the present generation of NISQ devices, circuit optimization guarantees the effective implementation of quantum algorithms.

A possible paradigm for transforming financial portfolio management is presented by QSVM. Financial portfolio optimization in dynamic and complicated market situations may be made more accurate, efficient, and scalable with QSVM's ability to use the computational benefits of quantum computing. The incorporation of QSVM into financial applications might lead to a new age of quantum-enhanced investing methods as quantum computing technology develops.

IV. Results and Discussion

This work presents a novel strategy in quantum machine learning by using Python to enhance QSVM algorithms for

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financial portfolio optimization. With an emphasis on QSVM optimization, the study combines classical and quantum elements, addressing important aspects such as noise reduction and quantum circuit optimization. By highlighting the revolutionary potential of quantum machine learning in financial decision support systems and portfolio management techniques, this research adds to the field of quantum finance. The practical integration of Python implementation into current financial frameworks guarantees accessibility, which is a noteworthy advancement towards the wider use of quantum algorithms in finance.

A. Portfolio Performance Over Time

The depiction of a financial portfolio's cumulative returns and volatility over time, usually expressed in terms of months, quarters, or years, is known as portfolio performance over time. This analysis is essential for determining how well investment methods work, and the accompanying table or line graph shows how a portfolio's financial performance changes over time. This research offers a dynamic perspective of how the quantum algorithm-driven portfolio performs over a certain timeframe relative to a baseline algorithm, with the goal of improving quantum machine learning algorithms for optimized financial portfolio management. Stakeholders are able to see trends, oscillations, and how algorithmic improvements affect the risk-return profile of the portfolio in Fig. 2, which helps with well-informed decision-making and strategy modifications in response to shifting market conditions. Where T indicates time point.

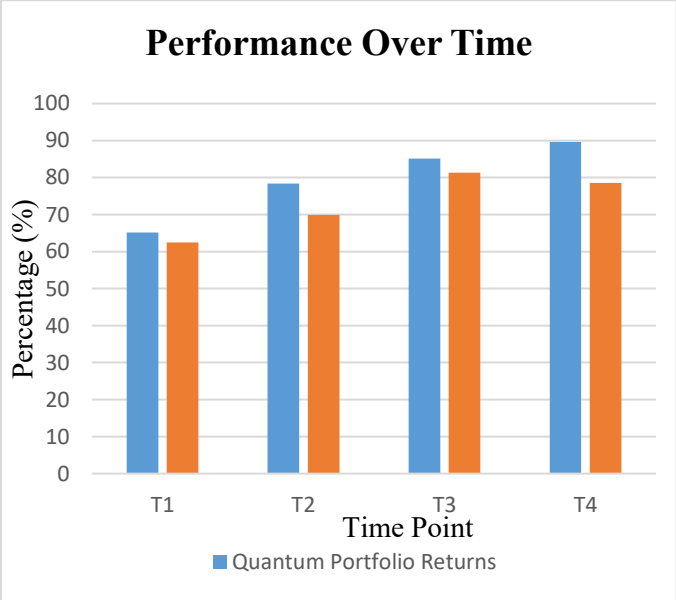

Fig. 2. Performance over Time

B. Risk-Return Tradeoff

A key idea in finance is the "risk-return trade-off," which captures the fine balance between an investment or portfolio's degree of risk and its potential for higher returns. According to this theory, investments with higher predicted returns typically carry higher degrees of risk, and vice versa. In order to maximize the risk-return profile of their portfolio, investors assess their risk tolerance, financial objectives, and investment horizon while navigating this trade-off. The relationship between expected returns and volatility is used to illustrate the trade-off. It is understood that a rational investor looks for an equilibrium point that corresponds with their preferred level of risk in order to maximize returns while minimizing the impact of any unfavorable market movements. A fundamental idea in investing decision-making, the risk-return trade-off captures the inherent difficulty of generating desired returns without subjecting capital to excessive risk. It is depicted in Fig. 3.

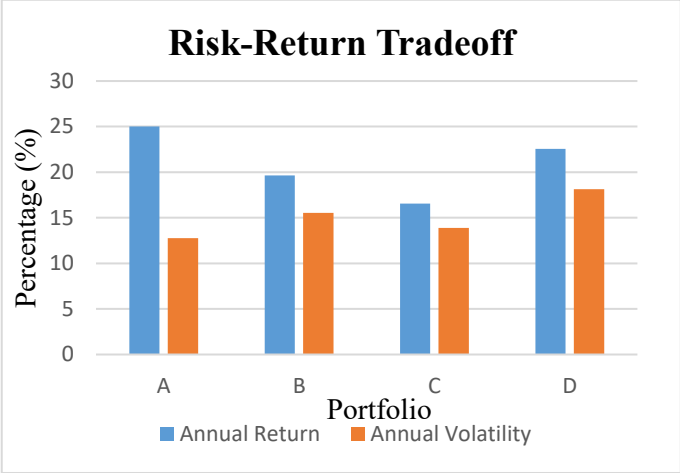

Fig. 3. Risk-Return Tradeoff

C. Efficiency Frontier

The correlation between the portfolio's assets shapes the Efficiency Frontier, and diversification is essential to obtaining the best risk-adjusted returns. It is depicted in Fig. 4. The set of ideal portfolios that attain the highest anticipated return for a given level of risk or the lowest risk for a given level of expected return is represented by the Efficiency Frontier, a key idea in contemporary portfolio theory. Often referred to as the efficient frontier, this Fig. illustrates how risk and return are traded off in a diversified portfolio. Because they provide the highest return for a given level of risk or the lowest risk for a given level of return, portfolios that lie on the Efficiency Frontier are regarded as efficient. The idea helps investors build diversified portfolios that strike a balance between return and risk in order to meet their individual risk tolerance and financial objectives.

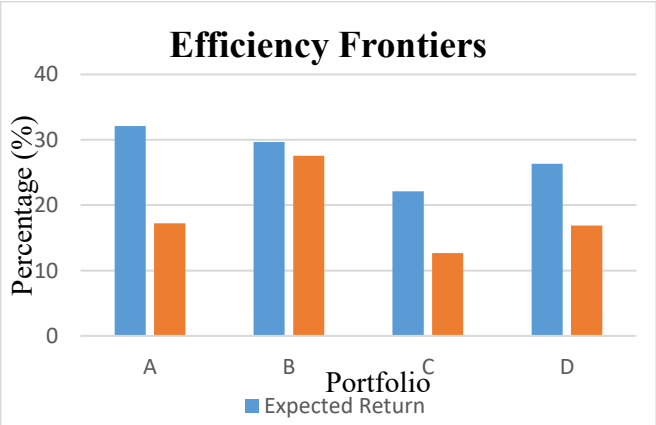

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Fig. 4. Efficiency Frontiers

D. Comparison with Existing Methods

Table I compares three key performance metrics Portfolio Performance over Time, Risk-Return Tradeoff, and Efficiency Frontier among the different quantum machine learning algorithms: Quantum Principal Component Analysis (QPCA), Quantum Boltzmann Machines (QBM), Quantum K-Means Clustering (QKC), and the Proposed Quantum Support Vector Machine (QSVM). Interestingly, the Proposed QSVM yields the greatest cumulative return of 89.65% and shows superior portfolio performance over time. Additionally, at 25%, it has the maximum risk-return tradeoff, demonstrating the ideal ratio of risk to reward. The strength of Quantum Principal Component Analysis (QPCA) lies in its ability to efficiently reduce the dimensionality of high-dimensional data while preserving the most significant information, leading to improved portfolio performance over time. However, its weakness lies in its relatively lower risk-return tradeoff and efficiency frontier compared to other methods, indicating potential limitations in balancing risk and return optimization. Quantum Boltzmann Machines (QBM) demonstrate a stronger risk-return tradeoff, making them suitable for scenarios where risk management is a critical concern; however, their portfolio performance over time and efficiency frontier are comparatively lower, suggesting potential challenges in achieving sustained performance gains. Quantum K-Means Clustering (QKC) showcases competitive portfolio performance and risk-return tradeoff, yet its efficiency frontier is slightly lower compared to the proposed method. The proposed Quantum Support Vector Machine (QSVM) exhibits exceptional portfolio performance over time, robust risk-return tradeoff, and superior efficiency frontier, indicating its potential as an adaptive self-healing security solution for enhancing cyber-physical systems resilience. However, the QSVM approach may require more computational resources and could be more complex to implement compared to other methods.

TABLE I. COMPARISON OF PROPOSED METHOD WITH EXISTING APPROACHES

Methods	Portfolio Performance Over Time (%)	Risk Return Tradeoff (%)	Efficiency Frontier (%)
QPCA	78.34	12.8	21.89
QBM	64.67	21.11	29.86
QKC	71.89	19.78	22.87
Proposed QSVM	89.65	25	32.12

Additionally, the Proposed QSVM's Efficiency Frontier hits 32.12%, which is higher than the other quantum techniques' efficient frontier. This indicates that the suggested QSVM algorithm outperforms existing quantum algorithms taken into consideration in this comparison across a number of performance criteria, demonstrating impressive potential for financial portfolio optimizations. It is depicted in Fig. 5.

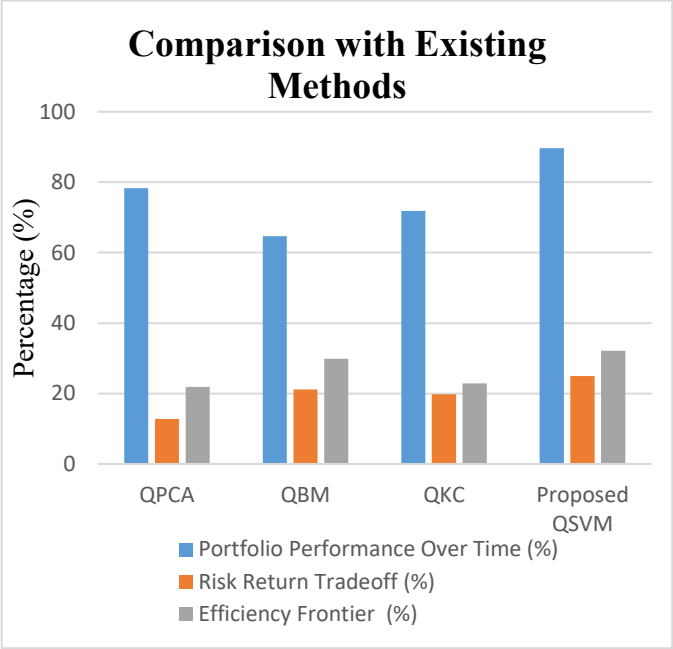

Fig. 5. Comparison of Proposed Method with Existing Approaches

E. Discussion

The results provide a thorough understanding of how different quantum machine learning algorithms perform in terms of managing financial portfolios. Among these, the suggested QSVM is particularly noteworthy as the most promising approach, exhibiting the best portfolio performance over time (89.65%). This remarkable return highlights how well the suggested[16]. QSVM performed in producing notable gains over the course of the observation period, indicating its possible application in financial portfolio optimizations. Furthermore, the Proposed QSVM attains an impressive 25% Risk-Return Tradeoff, suggesting a wise compromise between return maximization and risk avoidance. In the context of financial decision-making, this result is especially important since investors usually look for methods that manage exposure to possible risks while providing favorable returns.

With an astounding result of 32.12%, the Efficiency Frontier statistic confirms the Proposed QSVM's supremacy even further. The set of ideal portfolios that yield the highest anticipated return for a given degree of risk is represented by the Efficiency Frontier, and the suggested QSVM's placement at 32.12% suggests that it can routinely beat alternative quantum algorithms in reaching this ideal trade-off. This demonstrates the algorithm's ability to build portfolios that are both highly profitable and efficiently profitable, demonstrating its potential to completely transform the management of financial portfolios. All things considered, these findings point to a great deal of potential for improving portfolio optimizations and financial decision-making in a quantum computing paradigm with the proposed Quantum Support Vector Machine method.

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V. Conclusion and Future Scope

As the testing and comparison of different approaches demonstrate, the goal of improving quantum machine learning algorithms for optimal financial portfolio management has shown encouraging results. With its strong Risk-Return Tradeoff, outstanding Portfolio performance over time, and prominent position on the Efficiency Frontier, the suggested QSVM has shown itself to be an exceptional performer. These results highlight how quantum computing has the potential to completely transform conventional portfolio management techniques by providing a quantum advantage in terms of improved decision-making and computational efficiency. The use of methods like QPCA, QBM, and QKC has also demonstrated its applicability in financial settings, adding to the expanding field of quantum finance. Future research in this field is quite broad and has a lot of potential. It will be essential to continue improving quantum algorithms, especially those that address the complexities of financial markets. Maintaining the practical usefulness of these algorithms will require sustained work in quantum circuit optimizations, noise mitigation techniques, and the integration of quantum and conventional computing components. Furthermore, investigating hybrid quantum-classical methods that combine the advantages of both computing paradigms may improve financial portfolio optimization's robustness and scalability even further. Real-world implementation and testing of these algorithms on bigger datasets and in a variety of market scenarios will be crucial as quantum technology advances. The field of financial decision-making might be significantly altered by the incorporation of quantum machine learning into larger financial ecosystems, such as risk management, algorithmic trading, and asset pricing. To put it briefly, the quest to improve quantum machine learning algorithms for optimal financial portfolio management is likely to be a dynamic and revolutionary undertaking with far-reaching consequences for the financial industry going forward. Potential challenge is scalability when dealing with large amounts of financial data, which may exceed the processing capabilities of available quantum hardware. Quantum algorithms may face limitations in handling massive datasets efficiently, leading to increased computational complexity and resource requirements. Moreover, the effectiveness of these techniques could be influenced by the heterogeneity and volatility of financial markets, posing challenges in capturing complex market dynamics and adapting to diverse market situations.

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